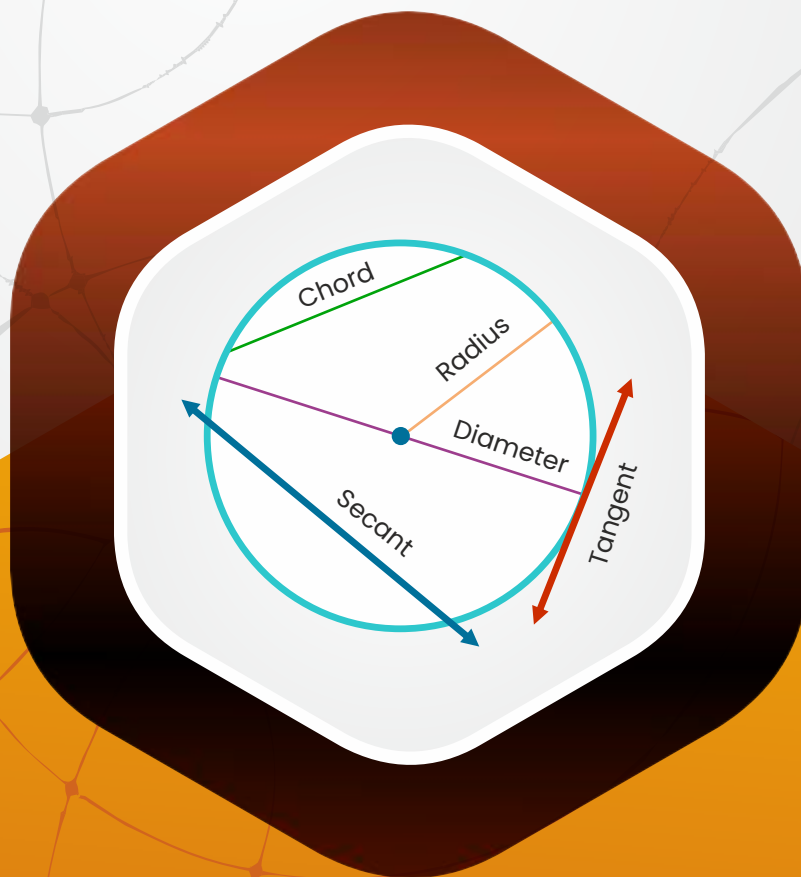


MATHEMATICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Study Material

Circle

(English Medium)

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CIRCLE

1. DEFINITION:

A circle is the locus of a point which moves in a plane in such a way that its distance from a fixed point (in the same given plane) remains constant. The fixed point is called the centre of the circle and the constant distance is called the radius of the circle.

Equation of a circle :

The curve traced by the moving point is called its circumference i.e. the equation of any circle is satisfied by co-ordinates of all points on its circumference.

or

The equation of the circle means the equation of its circumference.

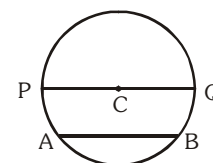
or

It is the set of all points lying on the circumference of the circle.

Chord and diameter - the line joining any two points on the circumference is called a chord. If any chord passing through its centre is called its diameter.

AB = chord, PQ = diameter

C = centre



2. STANDARD EQUATIONS OF THE CIRCLE :

(a) Central Form :

If (h, k) is the centre and r is the radius of the circle then its equation is

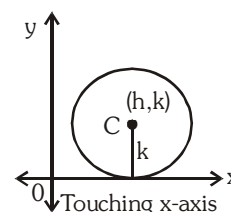
$$(x-h)^2 + (y-k)^2 = r^2$$

Special Cases :

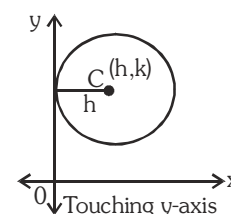
(i) If centre is origin $(0,0)$ and radius is ' r ' then equation of circle is $x^2 + y^2 = r^2$ and this is called the standard form.

(ii) If radius of circle is zero then equation of circle is $(x-h)^2 + (y-k)^2 = 0$. Such circle is called zero circle or **point circle**.

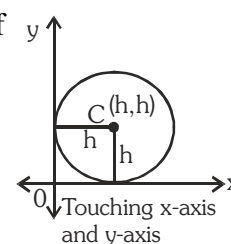
(iii) When circle touches x-axis then equation of the circle is $(x-h)^2 + (y-k)^2 = k^2$.



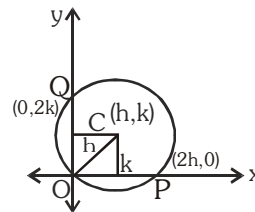
(iv) When circle touches y-axis then equation of circle is $(x-h)^2 + (y-k)^2 = h^2$.



(v) When circle touches both the axes (x-axis and y-axis) then equation of circle $(x-h)^2 + (y-h)^2 = h^2$.



- (vi) When circle passes through the origin and centre of the circle is (h, k) then radius $\sqrt{h^2 + k^2} = r$ and intercept cut on x-axis $OP = 2h$, and intercept cut on y-axis is $OQ = 2k$ and equation of circle is $(x-h)^2 + (y-k)^2 = h^2 + k^2$ or $x^2 + y^2 - 2hx - 2ky = 0$



Note : Centre of the circle may exist in any quadrant hence for general cases use $\pm$ sign before h & k .

(b) General equation of circle

$x^2 + y^2 + 2gx + 2fy + c = 0$. where g, f, c are constants and centre is $(-g, -f)$

i.e. $\left(-\frac{\text{coefficient of } x}{2}, -\frac{\text{coefficient of } y}{2} \right)$ and radius $r = \sqrt{g^2 + f^2 - c}$

Note :

- If $(g^2 + f^2 - c) > 0$, then r is real and positive and the circle is a real circle.
- If $(g^2 + f^2 - c) = 0$, then equation will represent a point.
- If $(g^2 + f^2 - c) < 0$, then r is imaginary then circle is also an imaginary circle with real centre.
- $x^2 + y^2 + 2gx + 2fy + c = 0$, has three constants and to get the equation of the circle at least three conditions should be known $\Rightarrow$ A unique circle passes through three non collinear points.
- The general second degree in x and y , $ax^2 + by^2 + 2hxy + 2gx + 2fy + c = 0$ represents a real circle if :**

coefficient of $x^2 =$ coefficient of y^2 or $a = b \neq 0$

and coefficient of $xy = 0$ or $h = 0$

and $(g^2 + f^2 - c) > 0$ (for a real circle)

$g^2 + f^2 - c = 0$ then it will represent a point.

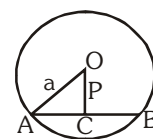
(c) Intercepts cut by the circle on axes :

The intercepts cut by the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ on :

- x-axis $= 2\sqrt{g^2 - c}$
- y-axis $= 2\sqrt{f^2 - c}$

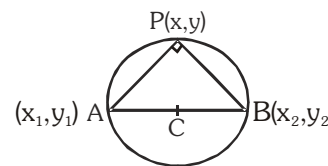
Note :

- If the circle cuts the x-axis at two distinct point, then $g^2 - c > 0$
- If the circle cuts the y-axis at two distinct point, then $f^2 - c > 0$
- If circle touches x-axis then $g^2 = c$.
- If circle touches y-axis then $f^2 = c$.
- Circle lies completely above or below the x-axis then $g^2 < c$.
- Circle lies completely to the right or left to the y-axis, then $f^2 < c$.
- Intercept cut by a line on the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ or length of chord of the circle $= 2\sqrt{a^2 - P^2}$ where a is the radius and P is the length of perpendicular from the centre to the chord.



(d) **Equation of circle in diameter form :**

If $A(x_1, y_1)$ and $B(x_2, y_2)$ are the end points of the diameter of the circle and $P(x, y)$ is the point other than A and B on the circle then from geometry we know that $\angle APB = 90^\circ$.



$$\Rightarrow (\text{Slope of PA}) \times (\text{Slope of PB}) = -1$$

$$\Rightarrow \therefore \left(\frac{y - y_1}{x - x_1} \right) \left(\frac{y - y_2}{x - x_2} \right) = -1$$

$$\Rightarrow (x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$$

Note : This will be the circle of least radius passing through (x_1, y_1) and (x_2, y_2)

(e) **Equation of circle in parametric forms :**

(i) The parametric equation of the circle $x^2 + y^2 = r^2$ are $x = r \cos \theta$, $y = r \sin \theta$; $\theta \in [0, 2\pi)$ and $(r \cos \theta, r \sin \theta)$ are called the parametric co-ordinates.

(ii) The parametric equation of the circle $(x - h)^2 + (y - k)^2 = r^2$ is $x = h + r \cos \theta$, $y = k + r \sin \theta$ where θ is parameter.

(iii) The parametric equation of the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ are

$$x = -g + \sqrt{g^2 + f^2 - c} \cos \theta,$$

$$y = -f + \sqrt{g^2 + f^2 - c} \sin \theta \text{ where } \theta \text{ is parameter.}$$

Note : Equation of a straight line joining two point α & β on the circle $x^2 + y^2 = a^2$ is

$$x \cos \frac{\alpha + \beta}{2} + y \sin \frac{\alpha + \beta}{2} = a \cos \frac{\alpha - \beta}{2}.$$

Illustration 1 : Find the centre and the radius of the circles

(a) $3x^2 + 3y^2 - 8x - 10y + 3 = 0$

(b) $x^2 + y^2 + 2x \sin \theta + 2y \cos \theta - 8 = 0$

(c) $2x^2 + \lambda xy + 2y^2 + (\lambda - 4)x + 6y - 5 = 0$, for some λ .

Solution :

(a) We rewrite the given equation as

$$x^2 + y^2 - \frac{8}{3}x - \frac{10}{3}y + 1 = 0 \Rightarrow g = -\frac{4}{3}, f = -\frac{5}{3}, c = 1$$

Hence the centre is $\left(\frac{4}{3}, \frac{5}{3} \right)$ and the radius is $\sqrt{\frac{16}{9} + \frac{25}{9} - 1} = \sqrt{\frac{32}{9}} = \frac{4\sqrt{2}}{3}$ units

(b) $x^2 + y^2 + 2x \sin \theta + 2y \cos \theta - 8 = 0$. Centre of this circle is $(-\sin \theta, -\cos \theta)$

Radius = $\sqrt{\sin^2 \theta + \cos^2 \theta + 8} = \sqrt{1 + 8} = 3$ units

(c) $2x^2 + \lambda xy + 2y^2 + (\lambda - 4)x + 6y - 5 = 0$

We rewrite the equation as

$$x^2 + \frac{\lambda}{2}xy + y^2 + \left(\frac{\lambda - 4}{2} \right)x + 3y - \frac{5}{2} = 0 \quad \dots\dots\dots (i)$$

Since, there is no term of xy in the equation of circle $\Rightarrow \frac{\lambda}{2} = 0 \Rightarrow \lambda = 0$

So, equation (i) reduces to $x^2 + y^2 - 2x + 3y - \frac{5}{2} = 0$

$\therefore$ centre is $\left(1, -\frac{3}{2} \right)$ Radius = $\sqrt{1 + \frac{9}{4} + \frac{5}{2}} = \frac{\sqrt{23}}{2}$ units.

Illustration 2 : If the lines $3x - 4y + 4 = 0$ and $6x - 8y - 7 = 0$ are tangents to a circle, then the radius of the circle is -

- (A) $3/2$ (B) $3/4$ (C) $1/10$ (D) $1/20$

Solution : The diameter of the circle is perpendicular distance between the parallel lines (tangents)

$$3x - 4y + 4 = 0 \text{ and } 3x - 4y - \frac{7}{2} = 0 \text{ and so it is equal to } \frac{4 + 7/2}{\sqrt{9+16}} = \frac{3}{2}.$$

$$\text{Hence radius is } \frac{3}{4}.$$

Ans. (B)

Illustration 3 : If $y = 2x + m$ is a diameter to the circle $x^2 + y^2 + 3x + 4y - 1 = 0$, then find m

Solution : Centre of circle $= (-3/2, -2)$. This lies on diameter $y = 2x + m$

$$\Rightarrow -2 = (-3/2) \times 2 + m \Rightarrow m = 1$$

Illustration 4 : The equation of a circle which passes through the point $(1, -2)$ and $(4, -3)$ and whose centre lies on the line $3x + 4y = 7$ is

- (A) $15(x^2 + y^2) - 94x + 18y - 55 = 0$ (B) $15(x^2 + y^2) - 94x + 18y + 55 = 0$
(C) $15(x^2 + y^2) + 94x - 18y + 55 = 0$ (D) none of these

Solution : Let the circle be $x^2 + y^2 + 2gx + 2fy + c = 0$ (i)

Hence, substituting the points, $(1, -2)$ and $(4, -3)$ in equation (i)

$$5 + 2g - 4f + c = 0 \text{ (ii)}$$

$$25 + 8g - 6f + c = 0 \text{ (iii)}$$

centre $(-g, -f)$ lies on line $3x + 4y = 7$

$$\text{Hence } -3g - 4f = 7$$

solving for g, f, c , we get

$$\text{Here } g = \frac{-47}{15}, f = \frac{9}{15}, c = \frac{55}{15}$$

$$\text{Hence the equation is } 15(x^2 + y^2) - 94x + 18y + 55 = 0$$

Ans. (B)

Illustration 5 : A circle has radius equal to 3 units and its centre lies on the line $y = x - 1$. Find the equation of the circle if it passes through $(7, 3)$.

Solution : Let the centre of the circle be (α, β) . It lies on the line $y = x - 1$

$$\Rightarrow \beta = \alpha - 1. \text{ Hence the centre is } (\alpha, \alpha - 1).$$

$$\Rightarrow \text{The equation of the circle is } (x - \alpha)^2 + (y - \alpha + 1)^2 = 9$$

$$\text{It passes through } (7, 3) \Rightarrow (7 - \alpha)^2 + (4 - \alpha)^2 = 9$$

$$\Rightarrow 2\alpha^2 - 22\alpha + 56 = 0 \Rightarrow \alpha^2 - 11\alpha + 28 = 0$$

$$\Rightarrow (\alpha - 4)(\alpha - 7) = 0 \Rightarrow \alpha = 4, 7$$

Hence the required equations are

$$x^2 + y^2 - 8x - 6y + 16 = 0 \text{ and } x^2 + y^2 - 14x - 12y + 76 = 0.$$

Ans.

Illustration 6 : Let L_1 be a straight line through the origin and L_2 be the straight line $x + y = 1$. If the intercepts made by the circle $x^2 + y^2 - x + 3y = 0$ on L_1 & L_2 are equal, then which of the following equations can represent L_1 ?

- (A) $x + y = 0$ (B) $x - y = 0$ (C) $x + 7y = 0$ (D) $x - 7y = 0$

Solution : Let L_1 be $y = mx$

lines L_1 & L_2 will be at equal distances from centre of the circle centre of the circle is

$$\left(\frac{1}{2}, -\frac{3}{2}\right)$$

$$\Rightarrow \frac{\left| \frac{1}{2}m + \frac{3}{2} \right|}{\sqrt{1+m^2}} = \frac{\left| \frac{1}{2} - \frac{3}{2} - 1 \right|}{\sqrt{2}} \Rightarrow \frac{(m+3)^2}{(1+m^2)} = 8$$

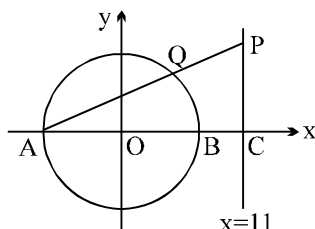
$$\Rightarrow 7m^2 - 6m - 1 = 0 \quad \Rightarrow (m-1)(7m+1) = 0$$

$$\Rightarrow m = 1, m = -\frac{1}{7} \quad \Rightarrow y = x, 7y + x = 0$$

Ans. (B, C)

Do yourself - 1 :

- Find the centre and radius of the circle $2x^2 + 2y^2 = 3x - 5y + 7$
- Find the equation to the circle
 - Whose radius is 10 and whose centre is $(-5, -6)$.
 - Whose radius is $a + b$ and whose centre is $(a, -b)$.
- Find the coordinates of the centres and the radii of the circles whose equations are :
 - $x^2 + y^2 - 4x - 8y = 41$
 - $\sqrt{1+m^2} (x^2 + y^2) - 2cx - 2mcy = 0$
- Find the equation of the circle whose centre is the point of intersection of the lines $2x - 3y + 4 = 0$ & $3x + 4y - 5 = 0$ and passes through the origin.
- Find the equation to the circle which touches the axis of :
 - x at a distance $+3$ from the origin and intercepts a distance 6 on the axis of y .
 - x , pass through the point $(1, 1)$ and have line $x + y = 3$ as diameter.
- Find the equation of the circle the end points of whose diameter are the centres of the circles $x^2 + y^2 + 16x - 14y = 1$ & $x^2 + y^2 - 4x + 10y = 2$
- The line $lx + my + n = 0$ intersects the curve $ax^2 + 2hxy + by^2 = 1$ at the point P and Q. The circle on PQ as diameter passes through the origin. Prove that $n^2(a + b) = l^2 + m^2$.
- Find the equation to the circle which goes through the origin and cuts off intercepts equal to h and k from the positive parts of the axes.
- Find the parametric form of the equation of the circle $x^2 + y^2 + px + py = 0$
- In the given figure, the circle $x^2 + y^2 = 25$ intersects the x -axis at the point A and B. The line $x = 11$ intersects the x -axis at the point C. Point P moves along the line $x = 11$ above the x -axis and AP intersects the circle at Q. Find



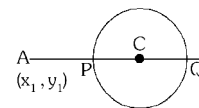
- The coordinates of the point P if the triangle AQB has the maximum area.
 - The coordinates of the point P if Q is the middle point of AP.
 - The coordinates of P if the area of the triangle AQB is $(1/4)^{\text{th}}$ of the area of the triangle APC.
- Find the equation to the circles which pass through the points :
 - $(0, 0)$, $(a, 0)$ and $(0, b)$
 - $(1, 2)$, $(3, -4)$ and $(5, -6)$
 - $(1, 1)$, $(2, -1)$ and $(3, 2)$

3. POSITION OF A POINT W.R.T CIRCLE :

(a) Let the circle is $x^2 + y^2 + 2gx + 2fy + c = 0$ and the point is (x_1, y_1) then -

Point (x_1, y_1) lies outside the circle or on the circle or inside the circle according as

$$\Rightarrow x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c >, =, < 0 \text{ or } S_1 >, =, < 0$$



(b) The greatest & the least distance of a point A from a circle with centre

C & radius r is $AC + r$ & $|AC - r|$ respectively.

4. POWER OF A POINT W.R.T. CIRCLE :

Theorem : The power of point $P(x_1, y_1)$ w.r.t. the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is S_1

$$\text{where } S_1 = x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$$

Note : If P outside, inside or on the circle then power of point is positive, negative or zero respectively.

If from a point $P(x_1, y_1)$, inside or outside the circle, a secant be drawn

intersecting the circle in two points A & B, then $PA \cdot PB = \text{constant}$.

The product $PA \cdot PB$ is called power of point $P(x_1, y_1)$ w.r.t. the circle

$$S \equiv x^2 + y^2 + 2gx + 2fy + c = 0, \text{ i.e. for number of secants } PA \cdot PB =$$

$$PA_1 \cdot PB_1 = PA_2 \cdot PB_2 = \dots = PT^2 = S_1$$

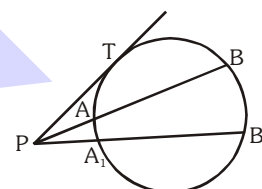


Illustration 7 : If $P(2, 8)$ is an interior point of a circle $x^2 + y^2 - 2x + 4y - p = 0$ which neither touches nor intersects the axes, then set for p is -

- (A) $p < -1$ (B) $p < -4$ (C) $p > 96$ (D) ϕ

Solution :

For internal point $p(2, 8)$, $4 + 64 - 4 + 32 - p < 0 \Rightarrow p > 96$

and x intercept $= 2\sqrt{1+p}$ therefore $1 + p < 0$

$$\Rightarrow p < -1 \text{ and y intercept } = 2\sqrt{4+p} \Rightarrow p < -4$$

Ans. (D)

Do yourself - 2 :

- Find the position of the points $(1, 2)$ & $(6, 0)$ w.r.t. the circle $x^2 + y^2 - 4x + 2y - 11 = 0$
- Find the greatest and least distance of a point $P(7, 3)$ from circle $x^2 + y^2 - 8x - 6y + 16 = 0$. Also find the power of point P w.r.t. circle.
- Determine the nature of the quadrilateral formed by four lines $3x + 4y - 5 = 0$; $4x - 3y - 5 = 0$; $3x + 4y + 5 = 0$ and $4x - 3y + 5 = 0$. Find the equation of the circle inscribed and circumscribing this quadrilateral.
- Find the shortest distance from the point $M(-7, 2)$ to the circle $x^2 + y^2 - 10x - 14y - 151 = 0$.
 - Find the co-ordinate of the point on the circle $x^2 + y^2 - 12x - 4y + 30 = 0$, which is farthest from the origin.

5. TANGENT LINE OF CIRCLE :

When a straight line meet a circle on two coincident points then it is called the tangent of the circle.

(a) Condition of Tangency :

The line $L = 0$ touches the circle $S = 0$ if P the length of the perpendicular from the centre to that line and radius of the circle r are equal i.e. $P = r$.

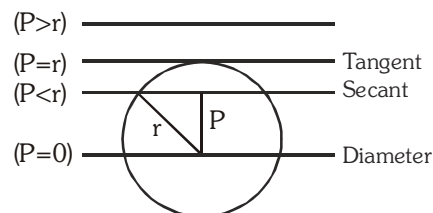


Illustration 8 : Find the range of parameter 'a' for which the variable line $y = 2x + a$ lies between the circles $x^2 + y^2 - 2x - 2y + 1 = 0$ and $x^2 + y^2 - 16x - 2y + 61 = 0$ without intersecting or touching either circle.

Solution : The given circles are $C_1 : (x - 1)^2 + (y - 1)^2 = 1$ and $C_2 : (x - 8)^2 + (y - 1)^2 = 4$

The line $y - 2x - a = 0$ will lie between these circle if centre of the circles lie on opposite sides of the line, i.e. $(1 - 2 - a)(1 - 16 - a) < 0 \Rightarrow a \in (-15, -1)$

Line wouldn't touch or intersect the circles if, $\frac{|1 - 2 - a|}{\sqrt{5}} > 1, \frac{|1 - 16 - a|}{\sqrt{5}} > 2$

$$\Rightarrow |1 + a| > \sqrt{5}, |15 + a| > 2\sqrt{5}$$

$$\Rightarrow a > \sqrt{5} - 1 \text{ or } a < -\sqrt{5} - 1, a > 2\sqrt{5} - 15 \text{ or } a < -2\sqrt{5} - 15$$

Hence common values of 'a' are $(2\sqrt{5} - 15, -\sqrt{5} - 1)$.

Illustration 9 : The equation of a circle whose centre is $(3, -1)$ and which cuts off a chord of length 6 on the line $2x - 5y + 18 = 0$

(A) $(x - 3)^2 + (y + 1)^2 = 38$

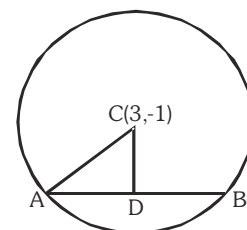
(B) $(x + 3)^2 + (y - 1)^2 = 38$

(C) $(x - 3)^2 + (y + 1)^2 = \sqrt{38}$

(D) none of these

Solution : Let $AB (= 6)$ be the chord intercepted by the line $2x - 5y + 18 = 0$ from the circle and let CD be the perpendicular drawn from centre $(3, -1)$ to the chord AB .

$$\text{i.e., } AD = 3, CD = \frac{2 \cdot 3 - 5(-1) + 18}{\sqrt{2^2 + 5^2}} = \sqrt{29}$$



$$\text{Therefore, } CA^2 = 3^2 + (\sqrt{29})^2 = 38$$

Hence required equation is $(x - 3)^2 + (y + 1)^2 = 38$

Ans. (A)

Illustration 10 : The area of the triangle formed by line joining the origin to the points of intersection(s) of the line $x\sqrt{5} + 2y = 3\sqrt{5}$ and circle $x^2 + y^2 = 10$ is

- (A) 3 (B) 4 (C) 5 (D) 6

Solution : Length of perpendicular from origin to the line $x\sqrt{5} + 2y = 3\sqrt{5}$ is

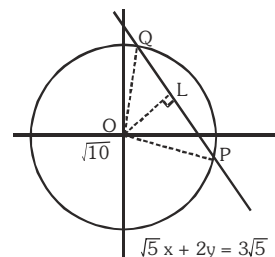
$$OL = \frac{3\sqrt{5}}{\sqrt{(\sqrt{5})^2 + 2^2}} = \frac{3\sqrt{5}}{\sqrt{9}} = \sqrt{5}$$

$$\text{Radius of the given circle} = \sqrt{10} = OQ = OP$$

$$PQ = 2QL = 2\sqrt{OQ^2 - OL^2} = 2\sqrt{10 - 5} = 2\sqrt{5}$$

$$\text{Thus area of } \triangle OPQ = \frac{1}{2} \times PQ \times OL = \frac{1}{2} \times 2\sqrt{5} \times \sqrt{5} = 5$$

Ans. (C)



(b) Equation of the tangent (T = 0) :

- (i) Tangent at the point (x_1, y_1) on the circle $x^2 + y^2 = a^2$ is $xx_1 + yy_1 = a^2$.
 (ii) (1) The tangent at the point $(a \cos t, a \sin t)$ on the circle $x^2 + y^2 = a^2$ is $x \cos t + y \sin t = a$
 (2) The point of intersection of the tangents at the points $P(\alpha)$ and $Q(\beta)$

$$\text{is } \left(\frac{a \cos \frac{\alpha + \beta}{2}}{\cos \frac{\alpha - \beta}{2}}, \frac{a \sin \frac{\alpha + \beta}{2}}{\cos \frac{\alpha - \beta}{2}} \right).$$

- (iii) The equation of tangent at the point (x_1, y_1) on the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$
 (iv) If line $y = mx + c$ is a straight line touching the circle $x^2 + y^2 = a^2$, then $c = \pm a\sqrt{1 + m^2}$ and contact points are $\left(\mp \frac{am}{\sqrt{1 + m^2}}, \pm \frac{a}{\sqrt{1 + m^2}} \right)$ or $\left(\mp \frac{a^2 m}{c}, \pm \frac{a^2}{c} \right)$ and equation of tangent is $y = mx \pm a\sqrt{1 + m^2}$
 (v) The equation of tangent with slope m of the circle $(x - h)^2 + (y - k)^2 = a^2$ is $(y - k) = m(x - h) \pm a\sqrt{1 + m^2}$

Note : To get the equation of tangent at the point (x_1, y_1) on any second degree curve we replace xx_1 in place of x^2 , yy_1 in place of y^2 , $\frac{x + x_1}{2}$ in place of x , $\frac{y + y_1}{2}$ in place of y , $\frac{xy_1 + yx_1}{2}$ in place of xy and c in place of c .

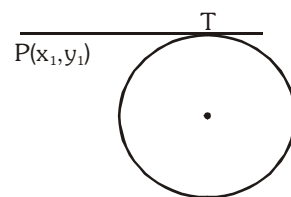
(c) Length of tangent ($\sqrt{S_1}$) :

The length of tangent drawn from point (x_1, y_1) outside the circle

$S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$ is,

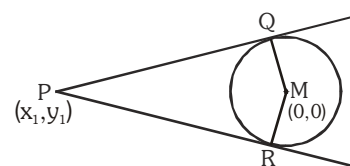
$$PT = \sqrt{S_1} = \sqrt{x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c}$$

Note : When we use this formula the coefficient of x^2 and y^2 must be 1.



(d) **Equation of Pair of tangents ($SS_1 = T^2$) :**

Let the equation of circle $S \equiv x^2 + y^2 = a^2$ and $P(x_1, y_1)$ is any point outside the circle. From the point we can draw two real and distinct tangent PQ & PR and combine equation of pair of tangents is -

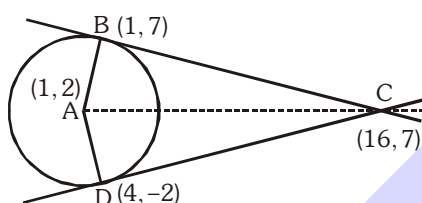


$$(x^2 + y^2 - a^2)(x_1^2 + y_1^2 - a^2) = (xx_1 + yy_1 - a^2)^2 \quad \text{or} \\ SS_1 = T^2$$

Illustration 11 : Let A be the centre of the circle $x^2 + y^2 - 2x - 4y - 20 = 0$ and B(1, 7) and D(4, -2) are points on the circle then, if tangents be drawn at B and D, which meet at C, then area of quadrilateral ABCD is -

- (A) 150 (B) 75 (C) 75/2 (D) none of these

Solution :



Here centre A(1, 2) and Tangent at (1, 7) is

$$x.1 + y.7 - 1(x + 1) - 2(y + 7) - 20 = 0 \text{ or } y = 7 \quad \dots\dots\dots (i)$$

$$\text{Tangent at D(4, -2) is } 3x - 4y - 20 = 0 \quad \dots\dots\dots (ii)$$

Solving (i) and (ii), C is (16, 7)

$$\text{Area ABCD} = AB \times BC = 5 \times 15 = 75 \text{ units.}$$

Ans. (B)

Do yourself - 3 :

- Find the equation of tangent to the circle $x^2 + y^2 - 2ax = 0$ at the point $(a(1 + \cos\alpha), a\sin\alpha)$.
- Find the equations of tangents to the circle $x^2 + y^2 - 6x + 4y - 12 = 0$ which are parallel to the line $4x - 3y + 6 = 0$
- Find the equation of the tangents to the circle $x^2 + y^2 = 4$ which are perpendicular to the line $12x - 5y + 9 = 0$. Also find the points of contact.
- Find the value of 'c' if the line $y = c$ is a tangent to the circle $x^2 + y^2 - 2x + 2y - 2 = 0$ at the point (1, 1)
- Write down the equation of the tangent to the circle $x^2 + y^2 - 3x + 10y = 15$ at the point (4, -11)
 - Find the condition that the straight line $3x + 4y = k$ may touch the circle $x^2 + y^2 = 10x$.
- Find the locus of the middle points of portions of the tangents to the circle $x^2 + y^2 = a^2$ terminated by the coordinate axes.

7. If M and m are the maximum and minimum values of $\frac{y}{x}$ for pair of real number (x, y) which satisfy the equation $(x - 3)^2 + (y - 3)^2 = 6$, then find the value of $(M + m)$.
8. Find the equation of the tangent to the circle
 - (a) $x^2 + y^2 - 6x + 4y = 12$, which are parallel to the straight line $4x + 3y + 5 = 0$.
 - (b) $x^2 + y^2 - 22x - 4y + 25 = 0$, which are perpendicular to the straight line $5x + 12y + 9 = 0$
 - (c) $x^2 + y^2 = 25$, which are inclined at 30° to the axis of x .
9. A circle passes through the points $(-1, 1)$, $(0, 6)$ and $(5, 5)$. Find the points on the circle the tangents at which are parallel to the straight line joining origin to the centre.
10. Tangents are drawn to the concentric circles $x^2 + y^2 = a^2$ and $x^2 + y^2 = b^2$ at right angle to one another. Show that the locus of their point of intersection is a 3rd concentric circle. Find its radius.
11. A variable circle passes through the point $A(a, b)$ & touches the x -axis. Show that the locus of the other end of the diameter through A is $(x - a)^2 = 4by$.
12. A circle is drawn with its centre on the line $x + y = 2$ to touch the line $4x - 3y + 4 = 0$ and pass through the point $(0, 1)$. Find its equation.
13. Show that the line $3x - 4y - c = 0$ will meet the circle having centre at $(2, 4)$ and the radius 5 in real and distinct points if $-35 < c < 15$.
14. Let L_1 be a straight line through the origin and L_2 be the straight line $x + y = 1$. If the intercepts made by the circle $x^2 + y^2 - x + 3y = 0$ on L_1 & L_2 are equal, then find the equation(s) which represent L_1 .
15. Find the equation of a line with gradient 1 such that the two circles $x^2 + y^2 = 4$ and $x^2 + y^2 - 10x - 14y + 65 = 0$ intercept equal length on it.
16. Find the equations of straight lines which pass through the intersection of the lines $x - 2y - 5 = 0$, $7x + y = 50$ & divide the circumference of the circle $x^2 + y^2 = 100$ into two arcs whose lengths are in the ratio 2 : 1.

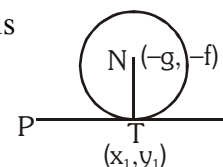
6. NORMAL OF CIRCLE :

Normal at a point is the straight line which is perpendicular to the tangent at the point of contact.

Note : Normal at point of the circle passes through the centre of the circle.

- (a) Equation of normal at point (x_1, y_1) of circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is

$$\mathbf{y} - \mathbf{y}_1 = \left(\frac{\mathbf{y}_1 + \mathbf{f}}{\mathbf{x}_1 + \mathbf{g}} \right) (\mathbf{x} - \mathbf{x}_1)$$



- (b) The equation of normal on any point (x_1, y_1) of circle $x^2 + y^2 = a^2$ is $\frac{y}{x} = \frac{y_1}{x_1}$.
- (c) If $x^2 + y^2 = a^2$ is the equation of the circle then at any point 't' of this circle $(a \cos t, a \sin t)$, the equation of normal is **$x \sin t - y \cos t = 0$** .

Illustration 12 : Find the equation of the normal to the circle $x^2 + y^2 - 5x + 2y - 48 = 0$ at the point $(5, 6)$.

Solution : Since normal to the circle always passes through the centre so equation of the normal will

be the line passing through $(5, 6)$ & $\left(\frac{5}{2}, -1\right)$

$$\text{i.e. } y + 1 = \frac{7}{5/2} \left(x - \frac{5}{2}\right) \Rightarrow 5y + 5 = 14x - 35$$

$$\Rightarrow 14x - 5y - 40 = 0$$

Ans.

Illustration 13 : If the straight line $ax + by = 2$; $a, b \neq 0$ touches the circle $x^2 + y^2 - 2x = 3$ and is normal to the circle $x^2 + y^2 - 4y = 6$, then the values of a and b are respectively

(A) 1, -1

(B) 1, 2

(C) $-\frac{4}{3}, 1$

(D) 2, 1

Solution :

Given $x^2 + y^2 - 2x = 3$

$\therefore$ centre is $(1, 0)$ and radius is 2

Given $x^2 + y^2 - 4y = 6$

$\therefore$ centre is $(0, 2)$ and radius is $\sqrt{10}$. Since line $ax + by = 2$ touches the first circle

$$\therefore \frac{|a(1) + b(0) - 2|}{\sqrt{a^2 + b^2}} = 2 \quad \text{or } |(a - 2)| = [2\sqrt{a^2 + b^2}] \quad \dots\dots\dots (i)$$

Also the given line is normal to the second circle. Hence it will pass through the centre of the second circle.

$$\therefore a(0) + b(2) = 2 \quad \text{or } 2b = 2 \quad \text{or } b = 1$$

Putting this value in equation (i) we get $|a - 2| = 2\sqrt{a^2 + 1^2}$ or $(a - 2)^2 = 4(a^2 + 1)$

$$\text{or } a^2 + 4 - 4a = 4a^2 + 4 \quad \text{or } 3a^2 + 4a = 0 \quad \text{or } a(3a + 4) = 0 \quad \text{or } a = 0, -\frac{4}{3} \quad (a \neq 0)$$

$$\therefore \text{ values of } a \text{ and } b \text{ are } \left(-\frac{4}{3}, 1\right).$$

Ans. (C)

Illustration 14 : Find the equation of a circle having the lines $x^2 + 2xy + 3x + 6y = 0$ as its normal and having size just sufficient to contain the circle $x(x - 4) + y(y - 3) = 0$.

Solution :

Pair of normals are $(x + 2y)(x + 3) = 0$

$\therefore$ Normals are $x + 2y = 0, x + 3 = 0$.

Point of intersection of normals is the centre of required circle i.e. $C_1(-3, 3/2)$ and centre

of given circle is $C_2(2, 3/2)$ and radius $r_2 = \sqrt{4 + \frac{9}{4}} = \frac{5}{2}$

Let r_1 be the radius of required circle

$$\Rightarrow r_1 = C_1C_2 + r_2 = \sqrt{(-3 - 2)^2 + \left(\frac{3}{2} - \frac{3}{2}\right)^2} + \frac{5}{2} = \frac{15}{2}$$

Hence equation of required circle is $x^2 + y^2 + 6x - 3y - 45 = 0$

Illustration 15 : The chord of contact of tangents drawn from a point on the circle $x^2 + y^2 = a^2$ to the circle $x^2 + y^2 = b^2$ touches the circle $x^2 + y^2 = c^2$. Show that a, b, c are in GP.

Solution : Let $P(a\cos\theta, a\sin\theta)$ be a point on the circle $x^2 + y^2 = a^2$.

Then equation of chord of contact of tangents drawn from

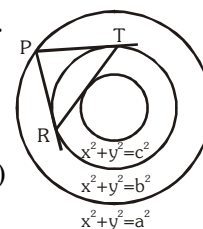
$P(a\cos\theta, a\sin\theta)$ to the circle $x^2 + y^2 = b^2$ is $ax\cos\theta + ay\sin\theta = b^2$ (i)

This touches the circle $x^2 + y^2 = c^2$ (ii)

$\therefore$ Length of perpendicular from $(0, 0)$ to (i) = radius of (ii)

$$\therefore \frac{|0 + 0 - b^2|}{\sqrt{(a^2 \cos^2 \theta + a^2 \sin^2 \theta)}} = c$$

or $b^2 = ac \Rightarrow a, b, c$ are in GP.



Do yourself - 5 :

- Find the equation of the chord of contact of the point $(1, 2)$ with respect to the circle $x^2 + y^2 + 2x + 3y + 1 = 0$
- Tangents are drawn from the point $P(4, 6)$ to the circle $x^2 + y^2 = 25$. Find the area of the triangle formed by them and their chord of contact.
- A point moving around circle $(x + 4)^2 + (y + 2)^2 = 25$ with centre C broke away from it either at the point A or point B on the circle and moved along a tangent to the circle passing through the point $D(3, -3)$. Find the following.
 - Equation of the tangents at A and B .
 - Coordinates of the points A and B .
 - Angle ADB and the maximum and minimum distances of the point D from the circle.
 - Area of quadrilateral $ADBC$ and the ΔDAB .
 - Equation of the circle circumscribing the ΔDAB and also the length of the intercepts made by this circle on the coordinate axes.
- Tangents OP and OQ are drawn from the origin O to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$. Find the equation of the circumcircle of the triangle OPQ .
- The straight line $x - 2y + 1 = 0$ intersects the circle $x^2 + y^2 = 25$ in points T and T' , find the co-ordinates of a point of intersection of tangents drawn at T and T' to the circle.

8. EQUATION OF THE CHORD WITH A GIVEN MIDDLE POINT ($T = S_1$) :

The equation of the chord of the circle $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$ in terms of its mid point M

(x_1, y_1) is $y - y_1 = -\frac{x_1 + g}{y_1 + f}(x - x_1)$. This on simplification can be put in the form

$xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$ which is designated by $T = S_1$.

Note that : The shortest chord of a circle passing through a point 'M' inside the circle, is one chord whose middle point is M.

Illustration 16 : Find the locus of middle points of chords of the circle $x^2 + y^2 = a^2$, which subtend right angle at the point $(c, 0)$.

Solution : Let $N(h, k)$ be the middle point of any chord AB , which subtend a right angle at $P(c, 0)$.

Since $\angle APB = 90^\circ$

$$\therefore NA = NB = NP$$

(since distance of the vertices from middle point of the hypotenuse are equal)

$$\text{or } (NA)^2 = (NB)^2 = (h - c)^2 + (k - 0)^2 \dots (i)$$

But also $\angle BNO = 90^\circ$

$$\therefore (OB)^2 = (ON)^2 + (NB)^2$$

$$\Rightarrow -(NB)^2 = (ON)^2 - (OB)^2 \Rightarrow -[(h - c)^2 + (k - 0)^2] = (h^2 + k^2) - a^2$$

$$\text{or } 2(h^2 + k^2) - 2ch + c^2 - a^2 = 0$$

$$\therefore \text{Locus of } N(h, k) \text{ is } 2(x^2 + y^2) - 2cx + c^2 - a^2 = 0$$

Ans.

Illustration 17 : Let a circle be given by $2x(x - a) + y(2y - b) = 0$ ($a \neq 0, b \neq 0$)

Find the condition on a and b if two chords, each bisected by the x -axis, can be drawn to the circle from $(a, b/2)$.

Solution : The given circle is $2x(x - a) + y(2y - b) = 0$

$$\text{or } x^2 + y^2 - ax - by/2 = 0$$

Let AB be the chord which is bisected by x -axis at a point M . Let its co-ordinates be $M(h, 0)$.

$$\text{and } S \equiv x^2 + y^2 - ax - by/2 = 0$$

$$\therefore \text{Equation of chord } AB \text{ is } T = S_1$$

$$hx + 0 - \frac{a}{2}(x + h) - \frac{b}{4}(y + 0) = h^2 + 0 - ah - 0$$

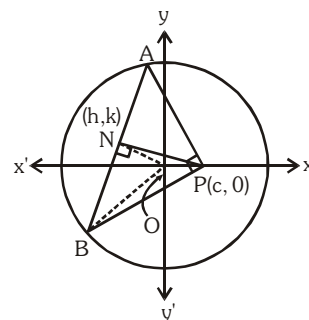
$$\text{Since it passes through } (a, b/2) \text{ we have } ah - \frac{a}{2}(a + h) - \frac{b^2}{8} = h^2 - ah \Rightarrow h^2 - \frac{3ah}{2} + \frac{a^2}{2} + \frac{b^2}{8} = 0$$

Now there are two chords bisected by the x -axis, so there must be two distinct real roots of h .

$$\therefore B^2 - 4AC > 0$$

$$\Rightarrow \left(\frac{-3a}{2}\right)^2 - 4 \cdot 1 \cdot \left(\frac{a^2}{2} + \frac{b^2}{8}\right) > 0 \Rightarrow a^2 > 2b^2.$$

Ans.



Do yourself - 6 :

- Find the equation of the chord of $x^2 + y^2 - 6x + 10 - a = 0$ which is bisected at $(-2, 4)$.
- Find the locus of mid point of chord of $x^2 + y^2 + 2gx + 2fy + c = 0$ that pass through the origin.
- Find the co-ordinates of the middle point of the chord which the circle $x^2 + y^2 - 2x + 2y - 2 = 0$ cuts off on the line $y = x - 1$.
Find also the equation of the locus of the middle point of all chords of the circle which are parallel to the line $y = x - 1$.
- Find the locus of the mid point of all chords of the circle $x^2 + y^2 - 2x - 2y = 0$ such that the pair of lines joining $(0, 0)$ & the point of intersection of the chords with the circles make equal angle with axis of x .

9. DIRECTOR CIRCLE :

The locus of point of intersection of two perpendicular tangents to a circle is called director circle. Let $P(h, k)$ is the point of intersection of two tangents drawn on the circle $x^2 + y^2 = a^2$. Then the equation of the pair of tangents is $SS_1 = T^2$

$$\text{i.e. } (x^2 + y^2 - a^2)(h^2 + k^2 - a^2) = (hx + ky - a^2)^2$$

As lines are perpendicular to each other then, coefficient of x^2 + coefficient of $y^2 = 0$

$$\Rightarrow [(h^2 + k^2 - a^2) - h^2] + [(h^2 + k^2 - a^2) - k^2] = 0$$

$$\Rightarrow h^2 + k^2 = 2a^2$$

$\therefore$ locus of (h, k) is $x^2 + y^2 = 2a^2$ which is the equation of the director circle.

$\therefore$ director circle is a concentric circle whose radius is $\sqrt{2}$ times the radius of the circle.

Note : The director circle of $x^2 + y^2 + 2gx + 2fy + c = 0$ is $x^2 + y^2 + 2gx + 2fy + 2c - g^2 - f^2 = 0$

Illustration 18 : Let P be any moving point on the circle $x^2 + y^2 - 2x = 1$, from this point chord of contact is drawn w.r.t. the circle $x^2 + y^2 - 2x = 0$. Find the locus of the circumcentre of the triangle CAB , C being centre of the circle and A, B are the points of contact.

Solution :

The two circles are

$$(x - 1)^2 + y^2 = 1 \quad \dots\dots\dots (i)$$

$$(x - 1)^2 + y^2 = 2 \quad \dots\dots\dots (ii)$$

So the second circle is the director circle of the first. So $\angle APB = \pi/2$

Also $\angle ACB = \pi/2$

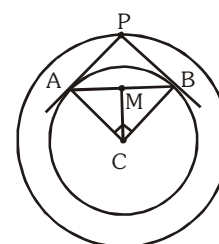
Now circumcentre of the right angled triangle CAB would lie on the mid point of AB

So let the point be $M \equiv (h, k)$

$$\text{Now, } CM = CB \sin 45^\circ = \frac{1}{\sqrt{2}}$$

$$\text{So, } (h - 1)^2 + k^2 = \left(\frac{1}{\sqrt{2}}\right)^2$$

$$\text{So, locus of } M \text{ is } (x - 1)^2 + y^2 = \frac{1}{2}.$$

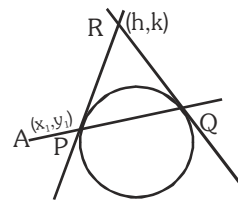


Do yourself - 7 :

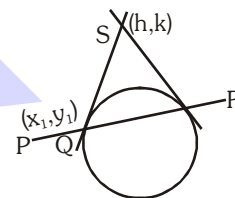
- Find the equation of the director circle of the circle $(x - h)^2 + (y - k)^2 = a^2$.
- If the angle between the tangents drawn to $x^2 + y^2 + 4x + 8y + c = 0$ from $(0, 0)$ is $\frac{\pi}{2}$, then find value of 'c'
- If two tangents are drawn from a point on the circle $x^2 + y^2 = 50$ to the circle $x^2 + y^2 = 25$, then find the angle between the tangents.

10. POLE AND POLAR :

Let any straight line through the given point $A(x_1, y_1)$ intersect the given circle $S=0$ in two points P and Q and if the tangent of the circle at P and Q meet at the point R then locus of point R is called polar of the point A and point A is called the pole, with respect to the given circle.

**(a) The equation of the polar of point (x_1, y_1) w.r.t. circle $x^2 + y^2 = a^2$ ($T=0$).**

Let PQR is a chord which passes through the point $P(x_1, y_1)$ which intersects the circle at points Q and R and the tangents are drawn at points Q and R meet at point $S(h, k)$ then equation of QR the chord of contact is $x_1 h + y_1 k = a^2$ $\therefore$ locus of point $S(h, k)$ is $xx_1 + yy_1 = a^2$ which is the equation of the polar.

**Note :**

- The equation of the polar is the $T=0$, so the polar of point (x_1, y_1) w.r.t circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$
- If point is outside the circle then equation of polar and chord of contact is same. So the chord of contact is polar.
- If point is inside the circle then chord of contact does not exist but polar exists.
- If point lies on the circle then polar, chord of contact and tangent on that point are same.
- If the polar of P w.r.t. a circle passes through the point Q, then the polar of point Q will pass through P and hence P & Q are conjugate points of each other w.r.t. the given circle.
- If pole of a line w.r.t. a circle lies on second line. Then pole of second line lies on first line and hence both lines are conjugate lines of each other w.r.t. the given circle.
- If O be the centre of a circle and P be any point, then OP is perpendicular to the polar of P.
- If O be the centre of a circle and P any point, then if OP (produce, if necessary) meet the polar of P in Q, then $OP \cdot OQ = (\text{radius})^2$

(b) Pole of a given line with respect to a circle

To find the pole of a line we assume the coordinates of the pole then from these coordinates we find the polar. This polar and given line represent the same line. Then by comparing the coefficients of similar terms we can get the coordinates of the pole. The pole of $\ell x + my + n = 0$

w.r.t. circle $x^2 + y^2 = a^2$ will be $\left(\frac{-\ell a^2}{n}, \frac{-ma^2}{n} \right)$

Centre is $\left(\frac{1}{1+\lambda}, \frac{2}{1+\lambda}\right)$ and radius = $\sqrt{\left(\frac{1}{1+\lambda}\right)^2 + \left(\frac{2}{1+\lambda}\right)^2 - \frac{1-\lambda}{1+\lambda}} = \frac{\sqrt{4+\lambda^2}}{|1+\lambda|}$.

Since it touches the line $x + 2y = 0$, hence

Radius = Perpendicular distance from centre to the line.

$$\text{i.e., } \left| \frac{\frac{1}{1+\lambda} + 2 \cdot \frac{2}{1+\lambda}}{\sqrt{1^2 + 2^2}} \right| = \frac{\sqrt{4+\lambda^2}}{|1+\lambda|} \Rightarrow \sqrt{5} = \sqrt{4+\lambda^2} \Rightarrow \lambda = \pm 1$$

$\lambda = -1$ cannot be possible in case of circle. So $\lambda = 1$.

Thus, we get the equation of circle.

Ans. (C)

Do yourself - 8 :

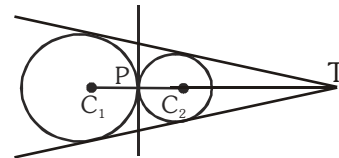
1. Prove that the polar of a given point with respect to any one of circles $x^2 + y^2 - 2kx + c^2 = 0$, where k is a variable, always passes through a fixed point, whatever be the value of k .
2. Find the equation of the circle through the points of intersection of the circles $x^2 + y^2 + 2x + 3y - 7 = 0$ and $x^2 + y^2 + 3x - 2y - 1 = 0$ and passing through the point $(1, 2)$.
3. Find the equation of the circle passing through the point of intersection of the circles $x^2 + y^2 - 6x + 2y + 4 = 0$, $x^2 + y^2 + 2x - 4y - 6 = 0$ and with its centre on the line $y = x$.
4. Find the equation of the circle passing through the points of intersection of the circles $x^2 + y^2 - 2x - 4y - 4 = 0$ and $x^2 + y^2 - 10x - 12y + 40 = 0$ and whose radius is 4.
5. Find the equation of the circle through points of intersection of the circle $x^2 + y^2 - 2x - 4y + 4 = 0$ and the line $x + 2y = 4$ which touches the line $x + 2y = 0$.
6. Find the equations of the circles which pass through the common points of the following pair of circles.
 - (a) $x^2 + y^2 + 2x + 3y - 7 = 0$ and $x^2 + y^2 + 3x - 2y - 1 = 0$ through the point $(1, 2)$
 - (b) $x^2 + y^2 + 4x - 6y - 12 = 0$ and $x^2 + y^2 - 5x + 17y = 19$ and having its centre on $x + y = 0$.
7. The line $2x - 3y + 1 = 0$ is tangent to a circle $S = 0$ at $(1, 1)$. If the radius of the circle is $\sqrt{13}$. Find the equation of the circle S .
8. Find the equation of the circle which passes through the point $(1, 1)$ & which touches the circle $x^2 + y^2 + 4x - 6y - 3 = 0$ at the point $(2, 3)$ on it.
9. Consider a family of circles passing through two fixed points $A(3, 7)$ & $B(6, 5)$. The chords in which the circle $x^2 + y^2 - 4x - 6y - 3 = 0$ cuts the members of the family are concurrent at a point. Find the coordinates of this point.

12. DIRECT AND TRANSVERSE COMMON TANGENTS :

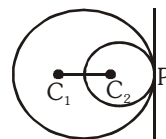
Let two circles having centre C_1 and C_2 and radii, r_1 and r_2 and C_1C_2 is the distance between their centres then :

(a) Both circles will touch :

- (i) **Externally** if $C_1C_2 = r_1 + r_2$ i.e. the distance between their centres is equal to sum of their radii and point P & T divides C_1C_2 in the ratio $r_1 : r_2$ (internally & externally respectively). In this case there are **three common tangents**.

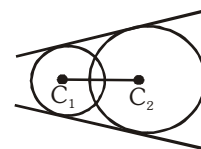


- (ii) **Internally** if $C_1C_2 = |r_1 - r_2|$ i.e. the distance between their centres is equal to difference between their radii and point P divides C_1C_2 in the ratio $r_1 : r_2$ **externally** and in this case there will be only **one common tangent**.



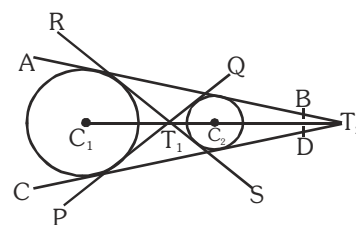
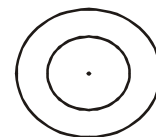
(b) The circles will intersect :

when $|r_1 - r_2| < C_1C_2 < r_1 + r_2$ in this case there are **two common tangents**.



(c) The circles will not intersect :

- (i) One circle will lie inside the other circle if $C_1C_2 < |r_1 - r_2|$ In this case there will be no common tangent.
- (ii) When circle are apart from each other then $C_1C_2 > r_1 + r_2$ and in this case there will be **four common tangents**. Lines PQ and RS are called **transverse** or **indirect** or **internal** common tangents and these lines meet line C_1C_2 on T_1 and T_1 divides the line C_1C_2 in the ratio $r_1 : r_2$ internally and lines AB & CD are called **direct** or **external** common tangents. These lines meet C_1C_2 produced on T_2 . Thus T_2 divides C_1C_2 externally in the ratio $r_1 : r_2$.



Note : Length of direct common tangent $= \sqrt{(C_1C_2)^2 - (r_1 - r_2)^2}$

Length of transverse common tangent $= \sqrt{(C_1C_2)^2 - (r_1 + r_2)^2}$

Illustration 20 : Prove that the circles $x^2 + y^2 + 2ax + c^2 = 0$ and $x^2 + y^2 + 2by + c^2 = 0$ touch each other,

$$\text{if } \frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{c^2}.$$

Solution : Given circles are $x^2 + y^2 + 2ax + c^2 = 0$ (i)
and $x^2 + y^2 + 2by + c^2 = 0$ (ii)

Let C_1 and C_2 be the centres of circles (i) and (ii), respectively and r_1 and r_2 be their radii, then

$$C_1 = (-a, 0), C_2 = (0, -b), r_1 = \sqrt{a^2 - c^2}, r_2 = \sqrt{b^2 - c^2}$$

Here we find the two circles touch each other internally or externally.

$$\text{For touch, } |C_1 C_2| = |r_1 \pm r_2|$$

$$\text{or } \sqrt{(a^2 + b^2)} = \left| \sqrt{(a^2 - c^2)} \pm \sqrt{(b^2 - c^2)} \right|$$

$$\text{On squaring } a^2 + b^2 = a^2 - c^2 + b^2 - c^2 \pm 2\sqrt{(a^2 - c^2)}\sqrt{(b^2 - c^2)}$$

$$\text{or } c^2 = \pm \sqrt{a^2 b^2 - c^2(a^2 + b^2) + c^4}$$

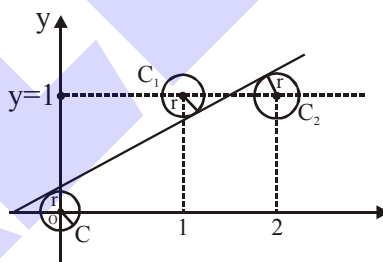
$$\text{Again squaring, } c^4 = a^2 b^2 - c^2(a^2 + b^2) + c^4$$

$$\text{or } c^2(a^2 + b^2) = a^2 b^2$$

$$\text{or } \frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{c^2}$$

Do yourself - 9 :

- Two circles with radius 5 touches at the point $(1, 2)$. If the equation of common tangent is $4x + 3y = 10$ and one of the circle is $x^2 + y^2 + 6x + 2y - 15 = 0$. Find the equation of other circle.
- Find the number of common tangents to the circles $x^2 + y^2 = 1$ and $x^2 + y^2 - 2x - 6y + 6 = 0$.
- As shown in the figure, three circles which have the same radius r , have centres at $(0,0)$; $(1,1)$ and $(2,1)$. If they have a common tangent line, as shown then, their radius ' r ' is -



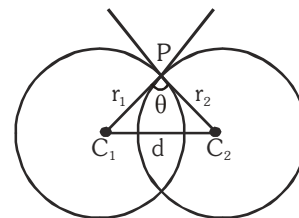
- (A) $\frac{\sqrt{5}-1}{2}$ (B) $\frac{\sqrt{5}}{10}$ (C) $\frac{1}{2}$ (D) $\frac{\sqrt{3}-1}{2}$
- Circle K is circle of largest radius, inscribed in the first quadrant touching the circle $x^2 + y^2 = 36$ internally. The length of the radius of the circle K, is-
(A) $\frac{6-\sqrt{2}}{2}$ (B) $\frac{3\sqrt{2}}{2}$ (C) 3 (D) $6(\sqrt{2}-1)$
 - A circle $S = 0$ is drawn with its centre at $(-1, 1)$ so as to touch the circle $x^2 + y^2 - 4x + 6y - 3 = 0$ externally. Find the intercept made by the circle $S = 0$ on the coordinate axes.
 - Find the equation of the circle whose radius is 3 and which touches the circle $x^2 + y^2 - 4x - 6y - 12 = 0$ internally at the point $(-1, -1)$.

13. THE ANGLE OF INTERSECTION OF TWO CIRCLES :

Definition : The angle between the tangents of two circles at the point of intersection of the two circles is called angle of intersection of two circles. If two circles are $S_1 \equiv x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$

$S_2 \equiv x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0$ and θ is the acute angle between them

$$\text{then } \cos \theta = \left| \frac{\left(\frac{r_1^2 + r_2^2 - d^2}{2r_1r_2} \right)}{\frac{2g_1g_2 + 2f_1f_2 - c_1 - c_2}{2\sqrt{g_1^2 + f_1^2 - c_1} \sqrt{g_2^2 + f_2^2 - c_2}}} \right|$$



Here r_1 and r_2 are the radii of the circles and d is the distance between their centres.

If the angle of intersection of the two circles is a right angle then such circles are called "**Orthogonal circles**" and conditions for the circles to be orthogonal is -

$$2g_1g_2 + 2f_1f_2 = c_1 + c_2$$

14. RADICAL AXIS OF THE TWO CIRCLES ($S_1 - S_2 = 0$) :

(a) **Definition :** The locus of a point, which moves in such a way that the length of tangents drawn from it to the circles are equal and is called the radical axis. If two circles are -

$$S_1 \equiv x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$$

$$S_2 \equiv x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0$$

Let $P(h,k)$ is a point and PA, PB are length of two tangents on the circles from point P , Then from definition -

$$\sqrt{h^2 + k^2 + 2g_1h + 2f_1k + c_1} = \sqrt{h^2 + k^2 + 2g_2h + 2f_2k + c_2} \text{ or } 2(g_1 - g_2)h + 2(f_1 - f_2)k + c_1 - c_2 = 0$$

$\therefore$ locus of (h,k)

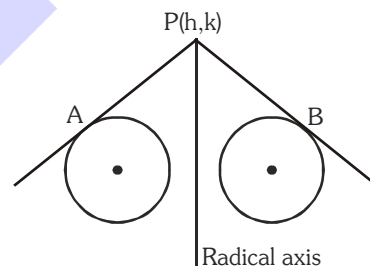
$$2x(g_1 - g_2) + 2y(f_1 - f_2) + c_1 - c_2 = 0$$

$$S_1 - S_2 = 0$$

which is the equation of radical axis.

Note :

- To get the equation of the radical axis first of all make the coefficient of x^2 and $y^2 = 1$
- If circles touch each other then radical axis is the common tangent to both the circles.
- When the two circles intersect on real points then common chord is the radical axis of the two circles.



- (iv) The radical axis of the two circles is perpendicular to the line joining the centre of two circles but not always pass through mid point of it.
- (v) Radical axis (if exist) bisects common tangent to two circles.
- (vi) The radical axes of three circles (taking two at a time) meet at a point.
- (vii) If circles are concentric then the radical axis does not exist but if circles are not concentric then radical axis always exists.
- (viii) If two circles are orthogonal to the third circle then radical axis of both circle passes through the centre of the third circle.
- (ix) A system of circle, every pair of which have the same radical axis, is called a **coaxial system of circles**.

(b) Radical centre :

The radical centre of three circles is the point from which length of tangents on three circles are equal i.e. the point of intersection of radical axis of the circles is the radical centre of the circles. To get the radical axis of three circles $S_1=0$, $S_2=0$, $S_3=0$ we have to solve any two

$$S_1 - S_2 = 0, S_2 - S_3 = 0, S_3 - S_1 = 0$$

Note :

- (i) The circle with centre as radical centre and radius equal to the length of tangent from radical centre to any of the circle, will cut the three circles orthogonally.
- (ii) If three circles are drawn on three sides of a triangle taking them as diameter then its orthocenter will be its radical centre.
- (iii) Locus of the centre of a variable circle orthogonal to two fixed circles is the radical axis between the two fixed circles.
- (iv) If two circles are orthogonal, then the polar of a point 'P' on first circle w.r.t. the second circle passes through the point Q which is the other end of the diameter through P. Hence locus of a point which moves such that its polars w.r.t. the circles $S_1 = 0$, $S_2 = 0$ & $S_3 = 0$ are concurrent is a circle which is orthogonal to all the three circles.

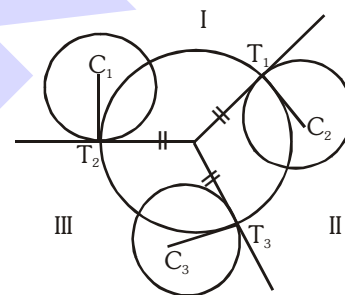


Illustration 21 : A and B are two fixed points and P moves such that $PA = nPB$ where $n \neq 1$. Show that locus of P is a circle and for different values of n all the circles have a common radical axis.

Solution :

Let $A \equiv (a, 0)$, $B \equiv (-a, 0)$ and $P(h, k)$

so $PA = nPB$

$$\Rightarrow (h - a)^2 + k^2 = n^2[(h + a)^2 + k^2]$$

$$\Rightarrow (1 - n^2)h^2 + (1 - n^2)k^2 - 2ah(1 + n^2) + (1 - n^2)a^2 = 0$$

$$\Rightarrow h^2 + k^2 - 2ah\left(\frac{1 + n^2}{1 - n^2}\right) + a^2 = 0$$

Hence locus of P is

$$x^2 + y^2 - 2ax \left(\frac{1+n^2}{1-n^2} \right) + a^2 = 0, \text{ which is a circle of different values of } n.$$

Let n_1 and n_2 are two different values of n so their radical axis is $x = 0$ i.e. y -axis. Hence for different values of n the circles have a common radical axis.

Illustration 22 : Find the equation of the circle through the points of intersection of the circles $x^2 + y^2 - 4x - 6y - 12 = 0$ and $x^2 + y^2 + 6x + 4y - 12 = 0$ and cutting the circle $x^2 + y^2 - 2x - 4 = 0$ orthogonally.

Solution : The equation of the circle through the intersection of the given circles is $x^2 + y^2 - 4x - 6y - 12 + \lambda(-10x - 10y) = 0$ (i)

where $(-10x - 10y = 0)$ is the equation of radical axis for the circle

$$x^2 + y^2 - 4x - 6y - 12 = 0 \text{ and } x^2 + y^2 + 6x + 4y - 12 = 0.$$

Equation (i) can be re-arranged as

$$x^2 + y^2 - x(10\lambda + 4) - y(10\lambda + 6) - 12 = 0$$

It cuts the circle $x^2 + y^2 - 2x - 4 = 0$ orthogonally.

$$\text{Hence } 2gg_1 + 2ff_1 = c + c_1$$

$$\Rightarrow 2(5\lambda + 2)(1) + 2(5\lambda + 3)(0) = -12 - 4 \Rightarrow \lambda = -2$$

Hence the required circle is

$$x^2 + y^2 - 4x - 6y - 12 - 2(-10x - 10y) = 0$$

$$\text{i.e., } x^2 + y^2 + 16x + 14y - 12 = 0$$

Illustration 23 : Find the radical centre of circles $x^2 + y^2 + 3x + 2y + 1 = 0$, $x^2 + y^2 - x + 6y + 5 = 0$ and $x^2 + y^2 + 5x - 8y + 15 = 0$. Also find the equation of the circle cutting them orthogonally.

Solution : Given circles are $S_1 \equiv x^2 + y^2 + 3x + 2y + 1 = 0$

$$S_2 \equiv x^2 + y^2 - x + 6y + 5 = 0$$

$$S_3 \equiv x^2 + y^2 + 5x - 8y + 15 = 0$$

$$\text{Equations of two radical axes are } S_1 - S_2 \equiv 4x - 4y - 4 = 0 \text{ or } x - y - 1 = 0$$

$$\text{and } S_2 - S_3 \equiv -6x + 14y - 10 = 0 \text{ or } 3x - 7y + 5 = 0$$

Solving them the radical centre is $(3, 2)$. Also, if r is the length of the tangent drawn from the radical centre $(3, 2)$ to any one of the given circles, say S_1 , we have

$$r = \sqrt{S_1} = \sqrt{3^2 + 2^2 + 3.3 + 2.2 + 1} = \sqrt{27}$$

Hence $(3, 2)$ is the centre and $\sqrt{27}$ is the radius of the circle intersecting them orthogonally.

$$\therefore \text{ Its equation is } (x - 3)^2 + (y - 2)^2 = r^2 = 27 \Rightarrow x^2 + y^2 - 6x - 4y - 14 = 0$$

Alternative Method :

Let $x^2 + y^2 + 2gx + 2fy + c = 0$ be the equation of the circle cutting the given circles orthogonally.

$$\therefore 2g\left(\frac{3}{2}\right) + 2f(1) = c + 1 \text{ or } 3g + 2f = c + 1 \text{ (i)}$$

$$2g\left(-\frac{1}{2}\right) + 2f(3) = c + 5 \quad \text{or} \quad -g + 6f = c + 5 \quad \dots\dots (ii)$$

$$\text{and} \quad 2g\left(\frac{5}{2}\right) + 2f(-4) = c + 15 \quad \text{or} \quad 5g - 8f = c + 15 \quad \dots\dots (iii)$$

Solving (i), (ii) and (iii) we get $g = -3$, $f = -2$ and $c = -14$

$\therefore$ equation of required circle is $x^2 + y^2 - 6x - 4y - 14 = 0$

Ans.

Do yourself - 10 :

- Find the angle of intersection of two circles
 $S : x^2 + y^2 - 4x + 6y + 11 = 0$ & $S' : x^2 + y^2 - 2x + 8y + 13 = 0$
- Find the equation of the radical axis of the circle $x^2 + y^2 - 3x - 4y + 5 = 0$ and $3x^2 + 3y^2 - 7x - 8y + 11 = 0$
- Find the radical centre of three circles described on the three sides $4x - 7y + 10 = 0$, $x + y - 5 = 0$ and $7x + 4y - 15 = 0$ of a triangle as diameters.
- The angle at which the circle $(x-1)^2 + y^2 = 10$ and $x^2 + (y-2)^2 = 5$ intersect is -
 (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$
- Find the radical centre of the following set of circles
 $x^2 + y^2 - 3x - 6y + 14 = 0$; $x^2 + y^2 - x - 4y + 8 = 0$; $x^2 + y^2 + 2x - 6y + 9 = 0$
- Find the equation of a circle which is co-axial with circles $2x^2 + 2y^2 - 2x + 6y - 3 = 0$ & $x^2 + y^2 + 4x + 2y + 1 = 0$. It is given that the centre of the circle to be determined lies on the radical axis of these two circles.
- Write the condition so that circles $x^2 + y^2 + 2ax + c = 0$ and $x^2 + y^2 + 2by + c = 0$ touch externally.

15. SOME IMPORTANT RESULTS TO REMEMBER :

- If the circle $S_1 = 0$, bisects the circumference of the circle $S_2 = 0$, then their common chord will be the diameter of the circle $S_2 = 0$.
- The radius of the director circle of a given circle is $\sqrt{2}$ times the radius of the given circle.
- The locus of the middle point of a chord of a circle subtend a right angle at a given point will be a circle.
- The length of side of an equilateral triangle inscribed in the circle $x^2 + y^2 = a^2$ is $\sqrt{3} a$
- If the lengths of tangents from the points A and B to a circle are ℓ_1 and ℓ_2 respectively, then if the points A and B are conjugate to each other, then $(AB)^2 = \ell_1^2 + \ell_2^2$.
- Length of transverse common tangent is less than the length of direct common tangent.

Do yourself - 11 :

- When the circles $x^2 + y^2 + 4x + 6y + 3 = 0$ and $2(x^2 + y^2) + 6x + 4y + c = 0$ intersect orthogonally, then find the value of c is
- Two circles whose radii are equal to 4 and 8 intersect at right angles. The length of their common chord is -
 (A) $\frac{16}{\sqrt{5}}$ (B) 8 (C) $4\sqrt{6}$ (D) $\frac{8\sqrt{5}}{5}$
- Two congruent circles with centres at (2,3) and (5,6) which intersect at right angles has radius equal to -
 (A) $2\sqrt{2}$ (B) 3 (C) 4 (D) none
- The equation of a circle which touches the line $x + y = 5$ at $N(-2,7)$ and cuts the circle $x^2 + y^2 + 4x - 6y + 9 = 0$ orthogonally, is -
 (A) $x^2 + y^2 + 7x - 11y + 38 = 0$ (B) $x^2 + y^2 = 53$
 (C) $x^2 + y^2 + x - y - 44 = 0$ (D) $x^2 + y^2 - x + y - 62 = 0$
- Find the equation to the circle orthogonal to the two circles $x^2 + y^2 - 4x - 6y + 11 = 0$; $x^2 + y^2 - 10x - 4y + 21 = 0$ and has $2x + 3y = 7$ as diameter.
- Find the equation of the circle through the points of intersection of circles $x^2 + y^2 - 4x - 6y - 12 = 0$ and $x^2 + y^2 + 6x + 4y - 12 = 0$ & cutting the circle $x^2 + y^2 - 2x - 4 = 0$ orthogonally.
- The centre of the circle $S = 0$ lie on the line $2x - 2y + 9 = 0$ & $S = 0$ cuts orthogonally the circle $x^2 + y^2 = 4$. Show that circle $S = 0$ passes through two fixed points & find their coordinates.
- (a) Find the equation of a circle passing through the origin if the line pair, $xy - 3x + 2y - 6 = 0$ is orthogonal to it. If this circle is orthogonal to the circle $x^2 + y^2 - kx + 2ky - 8 = 0$ then find the value of k .
 (b) Find the equation of the circle which cuts the circle $x^2 + y^2 - 14x - 8y + 64 = 0$ and the coordinate axes orthogonally.
- Find the equation to the circle, cutting orthogonally each of the following circles :
 $x^2 + y^2 - 2x + 3y - 7 = 0$; $x^2 + y^2 + 5x - 5y + 9 = 0$; $x^2 + y^2 + 7x - 9y + 29 = 0$.

Miscellaneous Illustrations :

Illustration 24 : Find the equation of a circle which passes through the point (2, 0) and whose centre is the limit of the point of intersection of the lines $3x + 5y = 1$ and $(2 + c)x + 5c^2y = 1$ as $c \rightarrow 1$.

Solution : Solving the equations $(2 + c)x + 5c^2y = 1$ and $3x + 5y = 1$

$$\text{then } (2 + c)x + 5c^2\left(\frac{1 - 3x}{5}\right) = 1 \quad \text{or} \quad (2 + c)x + c^2(1 - 3x) = 1$$

$$\therefore x = \frac{1-c^2}{2+c-3c^2} \quad \text{or} \quad x = \frac{(1+c)(1-c)}{(3c+2)(1-c)} = \frac{1+c}{3c+2}$$

$$\therefore x = \lim_{c \rightarrow 1} \frac{1+c}{3c+2} \quad \text{or} \quad x = \frac{2}{5}$$

$$\therefore y = \frac{1-3x}{5} = \frac{1-\frac{6}{5}}{5} = -\frac{1}{25}$$

Therefore the centre of the required circle is $\left(\frac{2}{5}, -\frac{1}{25}\right)$ but circle passes through $(2, 0)$

$$\therefore \text{Radius of the required circle} = \sqrt{\left(\frac{2}{5}-2\right)^2 + \left(-\frac{1}{25}-0\right)^2} = \sqrt{\frac{64}{25} + \frac{1}{625}} = \sqrt{\frac{1601}{625}}$$

Hence the required equation of the circle is $\left(x - \frac{2}{5}\right)^2 + \left(y + \frac{1}{25}\right)^2 = \frac{1601}{625}$

$$\text{or} \quad 25x^2 + 25y^2 - 20x + 2y - 60 = 0$$

Ans.

Illustration 25 : Two straight lines rotate about two fixed points. If they start from their position of coincidence such that one rotates at the rate double that of the other. Prove that the locus of their point of intersection is a circle.

Solution : Let $A \equiv (-a, 0)$ and $B \equiv (a, 0)$ be two fixed points.

Let one line which rotates about B an angle θ with the x-axis at any time t and at that time the second line which rotates about A make an angle 2θ with x-axis.

Now equation of line through B and A are respectively

$$y - 0 = \tan\theta(x - a) \quad \dots\dots (i)$$

$$\text{and} \quad y - 0 = \tan 2\theta(x + a) \quad \dots\dots (ii)$$

$$\text{From (ii),} \quad y = \frac{2 \tan \theta}{1 - \tan^2 \theta} (x + a)$$

$$= \left\{ \frac{\frac{2y}{(x-a)}}{1 - \frac{y^2}{(x-a)^2}} \right\} (x+a) \quad \text{(from (i))}$$

$$\Rightarrow y = \frac{2y(x-a)(x+a)}{(x-a)^2 - y^2} \Rightarrow (x-a)^2 - y^2 = 2(x^2 - a^2)$$

or $x^2 + y^2 + 2ax - 3a^2 = 0$ which is the required locus.

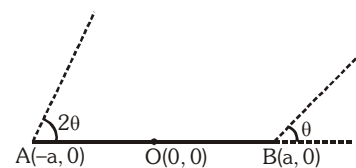


Illustration 26 : If the circle $x^2 + y^2 + 6x - 2y + k = 0$ bisects the circumference of the circle $x^2 + y^2 + 2x - 6y - 15 = 0$, then $k =$

- (A) 21 (B) -21 (C) 23 (D) -23

Solution : $2g_2(g_1 - g_2) + 2f_2(f_1 - f_2) = c_1 - c_2$
 $2(1)(3 - 1) + 2(-3)(-1 + 3) = k + 15$

$$4 - 12 = k + 15 \text{ or } -8 = k + 15 \Rightarrow k = -23$$

Ans. (D)

Illustration 27 : Find the equation of the circle of minimum radius which contains the three circles.

$$S_1 \equiv x^2 + y^2 - 4y - 5 = 0$$

$$S_2 \equiv x^2 + y^2 + 12x + 4y + 31 = 0$$

$$S_3 \equiv x^2 + y^2 + 6x + 12y + 36 = 0$$

Solution : For S_1 , centre = (0, 2) and radius = 3

For S_2 , centre = (-6, -2) and radius = 3

For S_3 , centre = (-3, -6) and radius = 3

let $P(a, b)$ be the centre of the circle passing through the centres of the three given circles, then

$$(a - 0)^2 + (b - 2)^2 = (a + 6)^2 + (b + 2)^2$$

$$\Rightarrow (a + 6)^2 - a^2 = (b - 2)^2 - (b + 2)^2$$

$$(2a + 6)6 = 2b(-4)$$

$$b = \frac{2 \times 6(a + 3)}{-8} = -\frac{3}{2}(a + 3)$$

$$\text{again } (a - 0)^2 + (b - 2)^2 = (a + 3)^2 + (b + 6)^2$$

$$\Rightarrow (a + 3)^2 - a^2 = (b - 2)^2 - (b + 6)^2$$

$$(2a + 3)3 = (2b + 4)(-8)$$

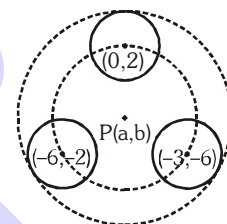
$$(2a + 3)3 = -16 \left[-\frac{3}{2}(a + 3) + 2 \right]$$

$$6a + 9 = -8(-3a - 5)$$

$$6a + 9 = 24a + 40$$

$$18a = -31$$

$$a = -\frac{31}{18}, b = -\frac{23}{12}$$



$$\text{radius of the required circle} = 3 + \sqrt{\left(-\frac{31}{18}\right)^2 + \left(-\frac{23}{12} - 2\right)^2} = 3 + \frac{5}{36}\sqrt{949}$$

$$\therefore \text{ equation of the required circle is } \left(x + \frac{31}{18}\right)^2 + \left(y + \frac{23}{12}\right)^2 = \left(3 + \frac{5}{36}\sqrt{949}\right)^2$$

Illustration 28 : Find the equation of the image of the circle $x^2 + y^2 + 16x - 24y + 183 = 0$ by the line mirror $4x + 7y + 13 = 0$.

Solution :

Centre of given circle = $(-8, 12)$, radius = 5

the given line is $4x + 7y + 13 = 0$

let the centre of required circle is (h, k)

since radius will not change. so radius of required circle is 5.

Now (h, k) is the reflection of centre $(-8, 12)$ in the line $4x + 7y + 13 = 0$

$$\text{Co-ordinates of A} = \left(\frac{-8+h}{2}, \frac{12+k}{2} \right)$$

$$\Rightarrow \frac{4(-8+h)}{2} + \frac{7(12+k)}{2} + 13 = 0$$

$$-32 + 4h + 84 + 7k + 26 = 0$$

$$4h + 7k + 78 = 0 \quad \dots\dots(i)$$

$$\text{Also } \frac{k-12}{h+8} = \frac{7}{4}$$

$$4k - 48 = 7h + 56$$

$$4k = 7h + 104 \quad \dots\dots(ii)$$

solving (i) & (ii)

$$h = -16, k = -2$$

$$\therefore \text{required circle is } (x + 16)^2 + (y + 2)^2 = 5^2$$

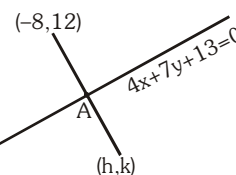


Illustration 29 : The circle $x^2 + y^2 - 6x - 10y + k = 0$ does not touch or intersect the coordinate axes and the point $(1, 4)$ is inside the circle. Find the range of the value of k .

Solution :

Since $(1, 4)$ lies inside the circle

$$\Rightarrow S_1 < 0$$

$$\Rightarrow (1)^2 + (4)^2 - 6(1) - 10(4) + k < 0$$

$$\Rightarrow k < 29$$

Also centre of given circle is $(3, 5)$ and circle does not touch or intersect the coordinate axes

$$\Rightarrow r < CA \quad \& \quad r < CB$$

$$CA = 5$$

$$CB = 3$$

$$\Rightarrow r < 5 \quad \& \quad r < 3$$

$$\Rightarrow r < 3 \quad \text{or} \quad r^2 < 9$$

$$r^2 = 9 + 25 - k$$

$$r^2 = 34 - k \quad \Rightarrow \quad 34 - k < 9$$

$$k > 25$$

$$\Rightarrow k \in (25, 29)$$

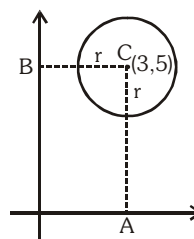


Illustration 30 : The circle $x^2 + y^2 - 4x - 8y + 16 = 0$ rolls up the tangent to it at $(2 + \sqrt{3}, 3)$ by 2 units, find the equation of the circle in the new position.

Solution : Given circle is $x^2 + y^2 - 4x - 8y + 16 = 0$

$$\text{let } P \equiv (2 + \sqrt{3}, 3)$$

Equation of tangent to the circle at $P(2 + \sqrt{3}, 3)$ will be

$$(2 + \sqrt{3})x + 3y - 2(x + 2 + \sqrt{3}) - 4(y + 3) + 16 = 0$$

or $\sqrt{3}x - y - 2\sqrt{3} = 0$

$$\text{slope} = \sqrt{3} \Rightarrow \tan\theta = \sqrt{3}$$

$$\theta = 60^\circ$$

line AB is parallel to the tangent at P

$\Rightarrow$ coordinates of point B = $(2 + 2\cos 60^\circ, 4 + 2\sin 60^\circ)$

thus $B = (3, 4 + \sqrt{3})$

$$\text{radius of circle} = \sqrt{2^2 + 4^2 - 16} = 2$$

$\therefore$ equation of required circle is $(x - 3)^2 + (y - 4 - \sqrt{3})^2 = 2^2$

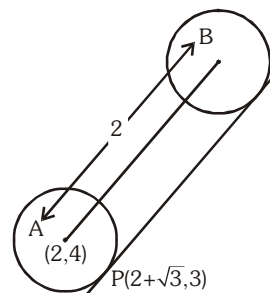


Illustration 31 : A fixed circle is cut by a family of circles all of which, pass through two given points $A(x_1, y_1)$ and $B(x_2, y_2)$. Prove that the chord of intersection of the fixed circle with any circle of the family passes through a fixed point.

Solution : Let $S = 0$ be the equation of fixed circle

let $S_1 = 0$ be the equation of any circle through A and B

which intersect $S = 0$ in two points.

$L \equiv S - S_1 = 0$ is the equation of the chord of intersection
of $S = 0$ and $S_1 = 0$

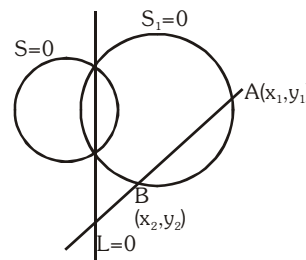
let $L_1 = 0$ be the equation of line AB

let S_2 be the equation of the circle whose diametrical ends are $A(x_1, y_1)$ & $B(x_2, y_2)$

then $S_1 \equiv S_2 - \lambda L_1 = 0$

$$\Rightarrow \quad L \equiv S - (S_0 - \lambda L_1) = 0 \quad \text{or} \quad L \equiv (S - S_0) + \lambda L_1 = 0$$

$$\text{or } L \equiv L' + \lambda L_1 = 0 \quad \dots\dots\dots(i)$$



(i) implies each chord of intersection passes through the fixed point, which is the point of intersection of lines $L' = 0$ & $L_1 = 0$. Hence proved.